

How Much Does Coldstone Charge?



Coldstone Creamery is an ice-cream store that makes hand-made ice cream and mixes in your choice of toppings on a frozen granite stone. Coldstone ice cream comes in three sizes: Like It™, Love It™, and Gotta Have It™.

- Below are the prices for a Love It™ size 1-topping ice cream order, and a Love It™ size 3-topping ice cream order.



Create Your Own Creation

Love It™
Strawberry Ice Cream
Roasted Almonds

\$4.79



Create Your Own Creation

Love It™
Chocolate Ice Cream
Chocolate Chips
Kit Kat®
M&M's®

\$6.57

Price per topping = $\frac{\Delta \text{Price}}{\Delta \# \text{ of toppings}}$

$$\frac{6.57 - 4.79}{2} = \$0.89$$

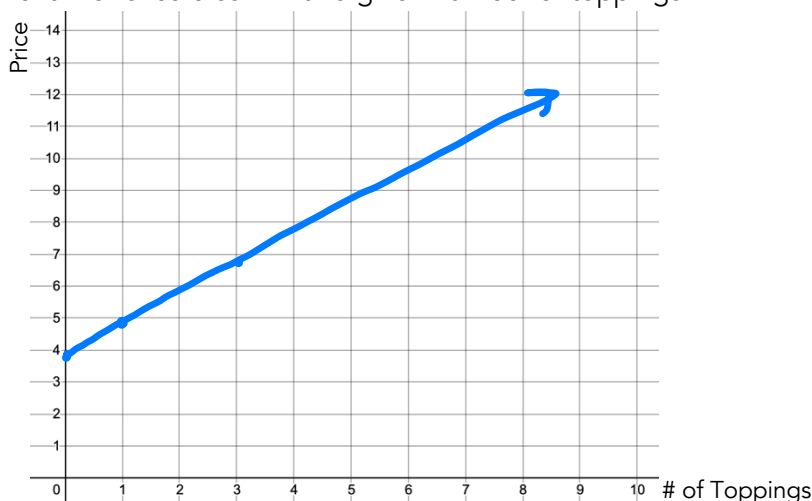
The difference in price represented 2 extra toppings.

- How much would you expect to pay for a Love It™ size bowl of ice cream with no toppings?

$$4.79 - 0.89 = \$3.90$$

- Write an equation for the cost of a Love It™ size ice cream with a given number of toppings and then graph it below.

$y = 3.90 + 0.89x$
where x is the # of toppings



- A Gotta Have It™ size ice cream with four toppings costs \$8.06. Assuming the price per topping is the same for all sizes, what is the price of a A Gotta Have It™ size ice cream with five toppings? Seven toppings? Clearly demonstrate your reasoning.

$$\$8.06 + 1(0.89) = \$8.95$$

$$\$8.06 + 3(0.89) = \$10.73$$

of additional toppings (7-4)

- Using your strategy in question 3, write an equation for the cost of a Gotta Have It™ size ice cream that does not require knowing the base price (no toppings) of the ice cream.

$$\$8.06 + \underbrace{(x-4)}_{\text{\# of additional toppings}}(0.89) = P$$

Pt. slope form

Lesson 0.5 – Linear Relationships

QuickNotes

- LT# 1: Linear relationships $\longleftrightarrow$ constant rate of change
- LT# 2 R.O.C = slope = $\frac{\Delta y}{\Delta x}$ "for every" "each"
- LT# 3: Point-slope form
 $m = \text{slope}$ (x_1, y_1) is ANY pt. on the line
 $y = y_1 + m(x - x_1)$ or $y - y_1 = m(x - x_1)$

Check Your Understanding

1. The cost of renting a U-Haul in Chicago is given by $C = 19.95 + 1.49m$ where m represents the number of miles driven.

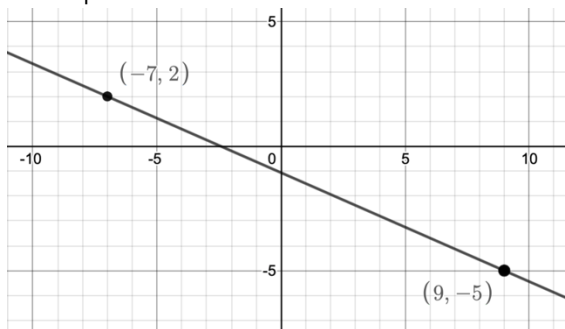
a. What does the 19.95 represent in the context of this problem?

base cost of renting, even when no miles are driven

b. What does the 1.49 represent in the context of this problem?

additional cost for every mile driven w/ the u-haul.

2. Write an equation for the line shown below.



$$\text{slope} = \frac{-5 - 2}{9 - (-7)} = \frac{-7}{16}$$

$$y - 2 = \frac{-7}{16}(x + 7)$$

3. A wedding venue charges a flat fee of \$900 and an additional fee of \$35 per guest attending. Write an equation that models this situation. If a couple has a budget of \$5000, how many guests can they invite?

$$C = 900 + 35x$$

$$5000 = 900 + 35x$$

$$4100 = 35x$$

$$x = 117.143$$

117 guests

4. In 2013, Netflix reported sales of 4.37 billion US Dollars and in 2016, this value increased to 8.83 billion dollars. Assuming that growth is linear, estimate the total sales in 2015 and the projected sales for 2020. Which estimate do you think is likely to be more accurate? Explain your reasoning.

$$\frac{8.83 - 4.37}{3} = 1.4867 \text{ billion/yr}$$

interpolation vs.

extrapolation

2015
2020

$$8.83 - 1.4867 = 7.34 \text{ billion}$$

$$8.83 + 4(1.4867) = 14.78 \text{ billion}$$

2015 is more reasonable bc it's within the data given.

Math Medic